



HR Management Platform

Work Breakdown Structure

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Project	HR Management Platform for Mid-Size Companies (Vietnam)
Client	Mid-size companies (50-500 employees)
Team	3 Developers + 1 Designer
Total Effort	259 man-days
Timeline	22-24 weeks (~5.5-6 months)
Mode	Estimate Mode (includes man-day estimates)

Executive Summary

This document presents the Work Breakdown Structure for an HR Management Platform targeting mid-size Vietnamese companies with 50 to 500 employees. The platform comprises five core modules: Employee Directory, Leave Management (compliant with Vietnam labor law), Payroll Integration with local Vietnamese providers, Performance Reviews, and Org Chart. The recommended approach is a modular monolith architecture using NestJS (backend), Next.js (web frontend), and React Native (mobile), with Auth0 for authentication and PostgreSQL for data persistence. Development is phased across three delivery milestones: MVP (Employee Directory, Leave Management, Org Chart, web only), Phase 2 (Performance Reviews, Mobile App), and Phase 3 (Payroll Integration). Total estimated effort is 259 man-days with a critical path timeline of 22-24 weeks for a team of 3 developers and 1 designer. The highest-risk item is payroll integration due to uncertain API availability from Vietnamese providers.

Technical Feasibility

Each major requirement has been assessed for technical feasibility. Items marked as needing a POC should be validated during the Discovery phase before committing to estimates.

Requirement	Feasibility	Assessment
Employee Directory	Viable	Standard CRUD with search. Well-understood pattern, highly automatable with Claude Code.
Leave Management	Viable	Core workflow is standard. Complexity lies in Vietnam labor law compliance (leave types, accrual rules). Requires legal review during discovery.
Payroll Integration	Needs POC	Vietnamese payroll providers vary widely in API maturity. Some may only support CSV import/export. A technical spike is required to evaluate top 3 providers before committing to integration approach.
Performance Reviews	Viable	Review cycles, goal tracking, and feedback collection are well-understood. Calibration views add moderate complexity but are not blocking.
Org Chart	Viable	Tree visualization libraries are mature. Drag-and-drop editing and real-time sync with directory are moderate effort.
Web Application	Viable	Next.js is a proven framework. Standard SPA with server-side rendering. No feasibility concerns.
Mobile Application	Viable	React Native enables code sharing with web. Team of 3 devs can manage if mobile scope is limited to core flows (directory, leave, notifications).
Vietnam Compliance (PDPD)	Needs POC	Vietnam's Personal Data Protection Decree has specific requirements for employee data handling, consent, and storage locality. Needs legal review.

Approach Analysis

Three architectural approaches were evaluated based on quality, time, and budget impact for a team of 3 developers and 1 designer building a multi-platform HR solution.

Criteria	A: Modular Monolith + React Native	B: Microservices + Flutter	C: SaaS + Customization
Architecture	NestJS modular monolith, Next.js web, React Native mobile, PostgreSQL	Separate NestJS services per domain, Flutter mobile, PostgreSQL per service	OrangeHRM/Odoo base, custom modules for VN-specific features
Quality	High. Single codebase simplifies testing and consistency. Shared logic between web and mobile.	High long-term, but inter-service complexity introduces integration risks for a small team.	Medium. Constrained by upstream platform decisions and customization limits.
Time Impact	Fastest. ~22-24 weeks. One backend to maintain, shared React ecosystem.	Slowest. ~35-40 weeks. DevOps overhead, two frontend stacks, service coordination.	Fast initially (~12 weeks for basic), but customization grows exponentially. ~30+ weeks total.
Budget Impact	Moderate. Auth0 licensing, AWS hosting. Predictable costs.	Highest. Multiple services = more infra, monitoring, and operational overhead.	Low initially, but licensing fees for SaaS platform + custom development on top.
Team Fit	Excellent. 3 devs can own the full stack. TypeScript throughout.	Poor. 3 devs cannot effectively manage multiple services + Flutter.	Fair. Less custom code, but requires platform expertise the team may not have.
AI Acceleration	Excellent. Claude Code for NestJS modules, React components, Prisma schemas, test generation.	Good. AI helps per-service, but coordination logic is harder to generate.	Limited. AI can help with custom modules but not with platform internals.
Recommendation	RECOMMENDED	Not recommended for this team size	Not recommended due to customization ceiling

WBS by SDLC Phase

1. Discovery & Requirements

Estimated effort: 15 man-days | Calendar: ~2 weeks

Module	Scope	Dependencies	Risk	Days
Requirements Documentation	Detailed functional specs and user stories for all 5 core modules. Acceptance criteria definition.	None	Low	5
Vietnam Labor Law Research	Research and document VN-specific leave policies, payroll rules, PDPD data handling requirements.	None	Medium	4
Payroll Provider Evaluation	Technical spike: identify top 3 VN payroll providers, assess API availability and integration feasibility.	None	High	4
UX Research & Wireframing	User flows, wireframes for all modules. Designer-led with stakeholder input.	Requirements Doc	Low	2

2. System Design & Architecture

Estimated effort: 12 man-days | Calendar: ~1.5 weeks

Module	Scope	Dependencies	Risk	Days
High-Level Architecture	System architecture diagram, service boundaries, deployment topology for NestJS modular monolith.	Discovery complete	Low	2
Data Model Design	PostgreSQL schema (Prisma) for employees, leave, reviews, org structure. Migration strategy.	Architecture	Low	3
API Contract Design	OpenAPI specs for all endpoints. Shared TypeScript types between web and mobile.	Data Model	Low	3
Security & Auth Architecture	Auth0 setup, RBAC design (Admin/HR/Manager/Employee roles), SSO integration plan.	Architecture	Medium	2
Infrastructure Blueprint	AWS infrastructure (ECS, RDS, S3, CloudFront), CI/CD pipeline design, environments.	Architecture	Low	2

3. Development — Phase 1 (MVP)

Estimated effort: 95 man-days | Calendar: ~8 weeks (3 devs parallel)

Module	Scope	Dependencies	Risk	Days
Auth & User Management	Auth0 integration, login/signup, RBAC middleware, session management, SSO.	Security Arch	Medium	10
Employee Directory	Employee profiles (CRUD), search/filter, profile photos, bulk import (CSV), i18n (VN/EN).	Auth Module	Low	12
Leave Management	VN labor law compliant leave types, request/approval workflows, balance tracking, calendar view, half-day support, public holidays.	Auth, Employee Dir	High	20
Org Chart	Visual hierarchy rendering, drag-and-drop editing, auto-sync with employee directory.	Employee Dir	Medium	10
Web Frontend (Core)	Next.js app shell, layout, navigation, dashboard, i18n framework, responsive design.	API Contracts	Low	15
Notification System	Email (SendGrid) + push (FCM) notifications for leave approvals and system alerts.	Auth Module	Low	5
Admin Panel	Company settings, user management, leave policy config, public holiday management.	Auth Module	Low	8
UI/UX Design (Full)	Complete design system, all screens, component library, design tokens. Designer-led.	Wireframes	Low	15

4. Development — Phase 2

Estimated effort: 55 man-days | Calendar: ~5 weeks

Module	Scope	Dependencies	Risk	Days
Performance Reviews	Review cycles, goal setting, self/peer/manager reviews, rating scales, templates, calibration view.	Employee Dir	Medium	20
Mobile App (React Native)	Core flows: employee directory, leave request/approval, notifications, profile. Shared logic with web.	Web Frontend, APIs	Medium	25
Reporting & Analytics	HR dashboards (headcount, leave utilization, review completion), CSV/PDF export.	All core modules	Low	10

5. Development — Phase 3 (Payroll)

Estimated effort: 30 man-days | Calendar: ~3 weeks

Module	Scope	Dependencies	Risk	Days
Payroll Integration	Integration with VN payroll providers, bidirectional data sync, reconciliation, error handling.	Payroll Spike, Leave Mgmt	High	25
Data Migration Tools	Import from Excel/CSV, data validation, dry-run mode, rollback capability.	Employee Dir	Medium	5

6. Quality Assurance

Estimated effort: 30 man-days | Calendar: ~3 weeks (overlaps with development)

Module	Scope	Dependencies	Risk	Days
Test Strategy & Setup	Jest + Playwright setup, CI integration, code coverage targets, AI-generated test boilerplate.	Dev start	Low	3
Unit & Integration Testing	Per-module testing (AI-assisted generation, human review), API contract tests.	Each module	Low	10
E2E Testing	Critical user journeys: leave request flow, review cycle, employee onboarding, payroll sync.	Feature complete	Low	7
UAT & Compliance Testing	User acceptance testing with VN HR stakeholders, Vietnam labor law compliance verification.	Feature complete	Medium	5
Security & Performance Testing	Penetration testing, load testing (500 concurrent users), PDPD compliance audit.	Feature complete	Medium	5

7. Deployment & DevOps

Estimated effort: 12 man-days | Calendar: ~1.5 weeks

Module	Scope	Dependencies	Risk	Days
CI/CD Pipeline	GitHub Actions, automated testing, staging/production deployment, Docker containerization.	Infra Blueprint	Low	3
Environment Setup	Staging + production environments on AWS, database provisioning, SSL, domain setup.	CI/CD	Low	3
Monitoring & Alerting	Application monitoring (CloudWatch/Datadog), error tracking (Sentry), uptime monitoring.	Deployment	Low	3
App Store Deployment	iOS App Store + Google Play Store submissions, certificates, review process.	Mobile App	Medium	3

8. Post-Launch & Maintenance

Estimated effort: 10 man-days | Calendar: ~1.5 weeks

Module	Scope	Dependencies	Risk	Days
Knowledge Transfer	Documentation, runbooks, training materials for client team.	All complete	Low	4
Bug Triage & Stabilization	Post-launch bug fixes, performance tuning during first 2 weeks.	Launch	Low	4
Phase 2+ Roadmap	Technical debt assessment, feature prioritization for next iteration.	Stabilization	Low	2

Dependency Map & Critical Path

The critical path determines the minimum timeline. Modules on the critical path cannot be parallelized and directly affect the total project duration.

Module	Depends On	Blocks	Critical Path
Requirements Documentation	—	Architecture, Wireframes	Yes
VN Labor Law Research	—	Leave Management	Yes
Payroll Provider Spike	—	Payroll Integration	No (parallel)
Architecture Design	Discovery	All development modules	Yes
Auth & User Management	Security Architecture	Employee Dir, Leave, Admin, Notifications	Yes
Employee Directory	Auth Module	Leave Mgmt, Org Chart, Reviews, Data Migration	Yes
Leave Management	Auth, Employee Dir, VN Law Research	Payroll Integration	Yes
Org Chart	Employee Directory	—	No
Performance Reviews	Employee Directory	—	No (Phase 2)
Mobile App	Web Frontend, APIs stable	App Store Deploy	No (Phase 2)
Payroll Integration	Payroll Spike, Leave Mgmt	—	No (Phase 3)

Critical path: Discovery → Architecture → Auth Module → Employee Directory → Leave Management → QA → Deployment. This sequence defines the minimum project duration of ~22 weeks.

Risk Register

Risk	Likelihood	Impact	Mitigation
Vietnamese payroll providers lack stable APIs, forcing CSV-based integration or custom adapters	High	High	Conduct payroll provider spike in Discovery. Design an adapter pattern that supports both API and CSV modes. Phase payroll integration last.
Vietnam labor law leave rules are more complex than anticipated, requiring rework	Medium	High	Engage a VN labor law consultant during Discovery. Build leave rules as configurable policies, not hardcoded logic.
PDPD compliance requires data localization or consent mechanisms not initially scoped	Medium	Medium	Include PDPD review in Discovery phase. Choose AWS ap-southeast-1 (Singapore) or VN-local hosting. Build consent flows into onboarding.
React Native mobile app has platform-specific bugs that delay release	Medium	Medium	Limit mobile MVP scope to core flows only. Use Expo for faster iteration. Build mobile in Phase 2 after web APIs stabilize.
Scope creep from client requesting additional features during development	High	Medium	Lock Phase 1 scope with signed-off requirements. Use phased delivery to defer additions to future phases.
Team capacity: 3 devs insufficient for parallel web + mobile development	Medium	Medium	Phase mobile after web MVP. Use AI tooling aggressively for boilerplate. Consider contract developer for mobile if timeline is rigid.
SSO integration complexity exceeds estimate due to varied client identity providers	Low	Medium	Use Auth0 which supports 30+ IdP integrations out of the box. Limit initial SSO to Google Workspace and Azure AD.

Timeline & Effort Summary

All estimates assume an AI-augmented team using Claude Code for boilerplate generation, AI-assisted test writing, and Dify workflow automation where applicable. Calendar weeks assume 3 developers working in parallel with overlapping phases.

Phase	Man-Days	Calendar Weeks	Start	End
Discovery & Requirements	15	2	Week 1	Week 2
System Design & Architecture	12	1.5	Week 3	Week 4
Development Phase 1 (MVP)	95	8	Week 5	Week 12
Development Phase 2	55	5	Week 13	Week 17
Development Phase 3 (Payroll)	30	3	Week 18	Week 20
Quality Assurance	30	3	Week 18	Week 20
Deployment & DevOps	12	1.5	Week 21	Week 22
Post-Launch & Maintenance	10	1.5	Week 23	Week 24

Total Effort	259 man-days
Team	3 developers + 1 designer
Critical Path Duration	22-24 weeks (~5.5-6 months)
MVP Delivery	Week 12 (web app with Employee Dir, Leave Mgmt, Org Chart)
Full Platform Delivery	Week 22-24 (all modules, web + mobile)

Assumptions & Constraints

Assumptions

- The team has access to AI tooling (Claude Code, Dify) and is proficient in their use, resulting in ~30-40% acceleration on boilerplate and testing tasks.
- Auth0 (or equivalent) is approved for authentication; the team will not build a custom auth system.
- Vietnamese payroll providers will cooperate during the evaluation spike and provide technical documentation.
- The client will provide timely feedback during UAT and requirements validation (max 3 business days turnaround).
- A Vietnam labor law consultant or HR domain expert is available during the Discovery phase.
- The designer works full-time on this project during Phases 1-2 and transitions out after the design system is delivered.
- AWS is the approved cloud provider; no on-premises hosting requirement.
- The mobile app scope is limited to core flows (directory, leave, notifications) and does not replicate the full web experience.

Constraints

- Team size is fixed at 3 developers + 1 designer. No additional hires are planned.
- Budget for third-party services (Auth0, SendGrid, AWS) must be approved separately.
- Payroll integration scope depends on the outcome of the provider evaluation spike; estimates may change.
- The platform must comply with Vietnam's Personal Data Protection Decree (PDPD), which may require data localization.
- All UI must support Vietnamese and English from launch; additional languages are out of scope.
- Performance targets: system must handle 500 concurrent users with sub-2-second page load times.